

프롬프트 하나로 데이터 분석 시키기: LangChain으로 배우는 AI 분석

LangChain - 개발 환경 준비

김종영



◆ Jupyter

- Jupyter 이름의 유래는 Julia+Python+R, ipython notebook 프로젝트로 시작
 - . Julia 프로그래밍 언어는 "과학 계산 목적의 빠른 고성능 언어"
 - . R 프로그래밍 언어는 "통계/데이터 분석 목적의 전문가용 도구"
- 오픈 소스, 여러가지 프로그래밍 언어를 지원, 인터랙티브 컴퓨팅 환경 제공
- 클라이언트 역할의 프론트엔드와 서버 역할의 백엔드로 구성
- 프론트엔드: 사용자가 마주하는 UI, 셀에 코드를 입력하고, 계산 결과나 그래프를 보여줌. 기본적으로 웹 브라우저에서 열리며 백엔드에 접속하는 형태
- 백엔드 (다른 말로 커널): 코드를 실제로 실행하는 엔진에 해당되며, 커널이 서버에서 실행이 되는 형태. 커널만 바꾸면 다양한 프로그래밍 언어를 Jupyter에서 실행 가능 → "여러가지 프로그래밍 언어를 지원"

◆ 파이썬 가상환경

- 분리된 파이썬 실행 환경
- Anaconda: conda+데이터 사이언스에 관련된 방대한 패키지
- miniconda: conda+필수적인 최소한의 라이브러리
- 가상 환경을 만들 때 사용하는 툴: venv, conda, pipenv, poetry
- **ipykernel이 설치된 파이썬 가상환경**을 VS Code에서 커널로 사용

miniconda, VS Code - 2

- `conda env list` # 모든 가상 환경 리스트
- `conda list` # 설치된 모든 패키지 리스트
- `conda create -n py310_64 python=3.10` # `py310_64`는 가상 환경 이름
- `conda activate py310_64` # 활성화
- `conda deactivate` # 비활성화

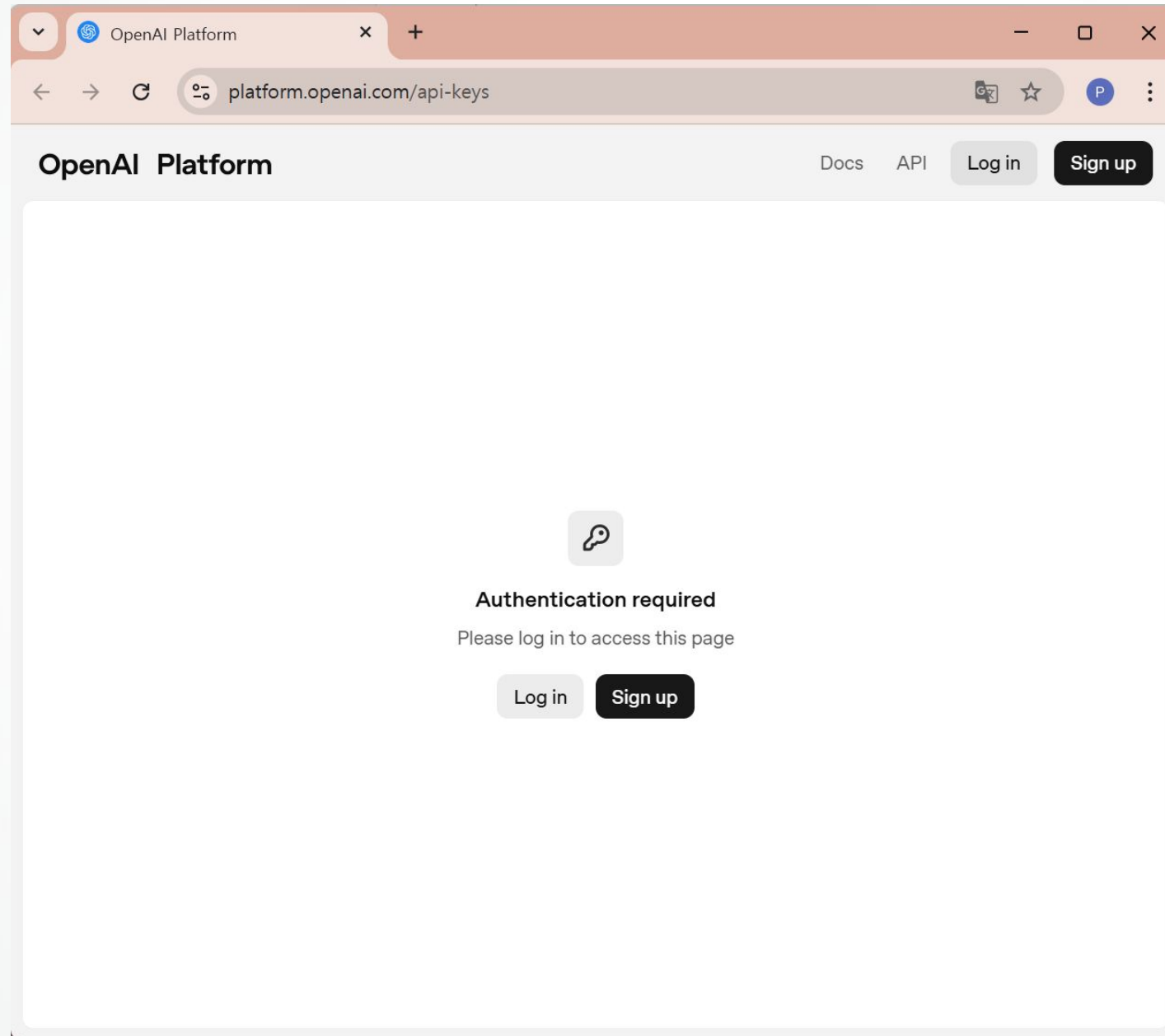
- `conda install ipykernel`
- `conda install langchain`
- `python -m pip install google-generativeai`
 - 현재 사용중인 파이썬 인터프리터에 연결된 pip를 실행

- conda는 미리 빌드된 바이너리 패키지를 설치하므로 빠르고 컴파일 관련 이슈가 없음
- pip는 소스 기반 설치이므로 컴파일 이슈가 있을 수 있음
- conda 설치를 추천하지만 Anaconda 저장소에 원하는 패키지가 없다면 pip로 설치해야 함

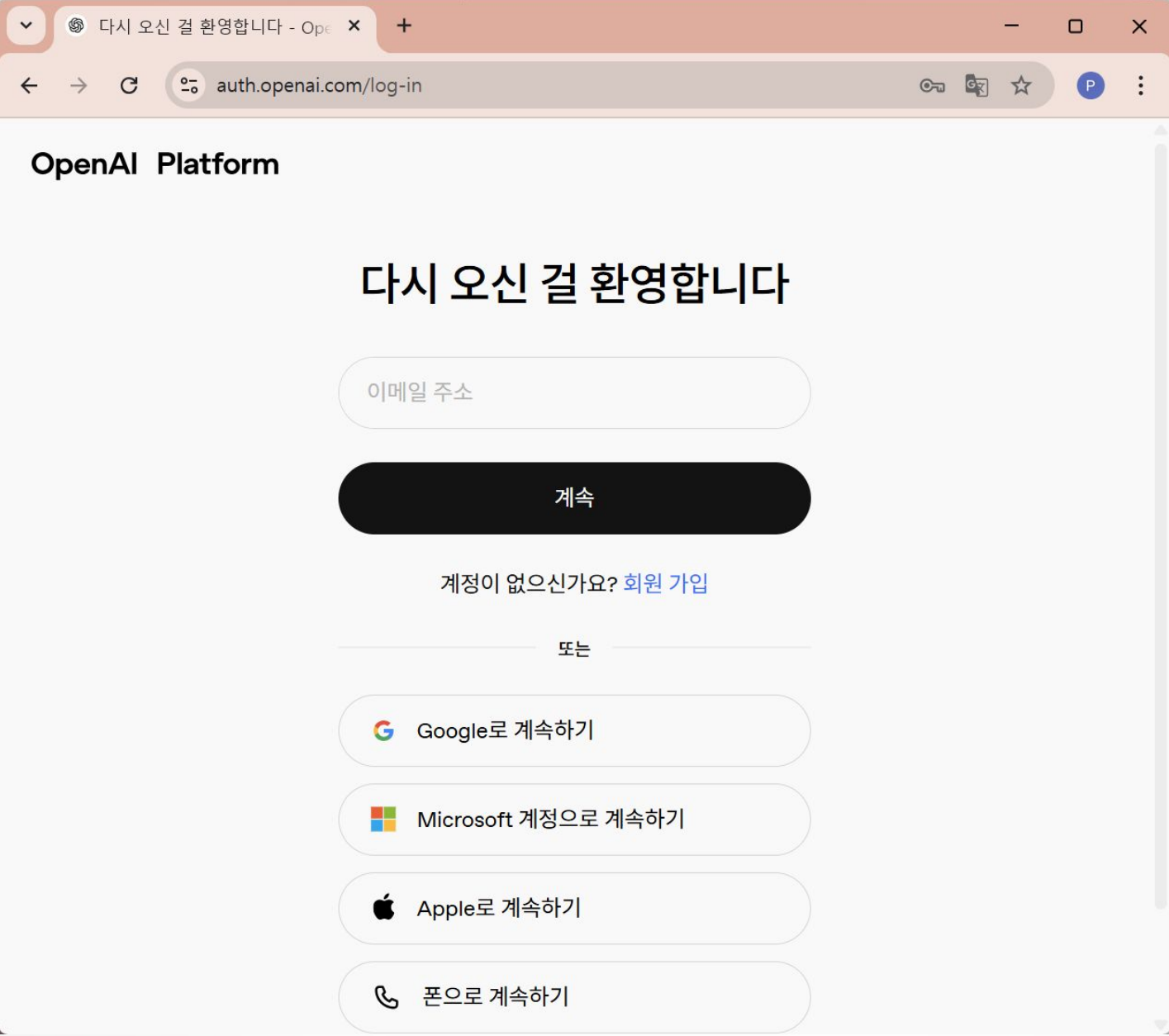
miniconda, VS Code - 2

- ① miniconda 설치 → <https://www.anaconda.com/download>
- ② VS Code 설치 → <https://code.visualstudio.com/>
- ③ conda 가상 환경 생성
- ④ VS Code에서 extension 설치
 - python (Microsoft): 자동완성, 코드 포매팅 (들여쓰기, 공백, 줄 바꿈), 디버깅
 - pylance (Microsoft): 정적 오류/경고 표시, 정의로 이동, 함수 시그니처 표시 등
 - jupyter (Microsoft): 셀 기반 UI, 원격 Jupyter 서버 연결 등

API Key 발급 - OpenAI(<https://platform.openai.com/api-keys>)



API Key 발급 - OpenAI(<https://platform.openai.com/api-keys>)



A screenshot of a web browser showing the OpenAI Platform login page. The browser's address bar displays 'auth.openai.com/log-in'. The page has a light gray background with the 'OpenAI Platform' logo at the top left. The main heading is '다시 오신 걸 환영합니다' (Welcome back). Below it is a text input field for '이메일 주소' (Email address). A large black button with the text '계속' (Continue) is positioned below the input field. Underneath the button, there is a link that says '계정이 없으신가요? 회원 가입' (Don't have an account? Sign up). A horizontal separator line with the word '또는' (or) in the center follows. Below the line are four rounded rectangular buttons for social login: 'Google로 계속하기' (Continue with Google), 'Microsoft 계정으로 계속하기' (Continue with Microsoft account), 'Apple로 계속하기' (Continue with Apple), and '폰으로 계속하기' (Continue with phone).

다시 오신 걸 환영합니다

이메일 주소

계속

계정이 없으신가요? [회원 가입](#)

또는

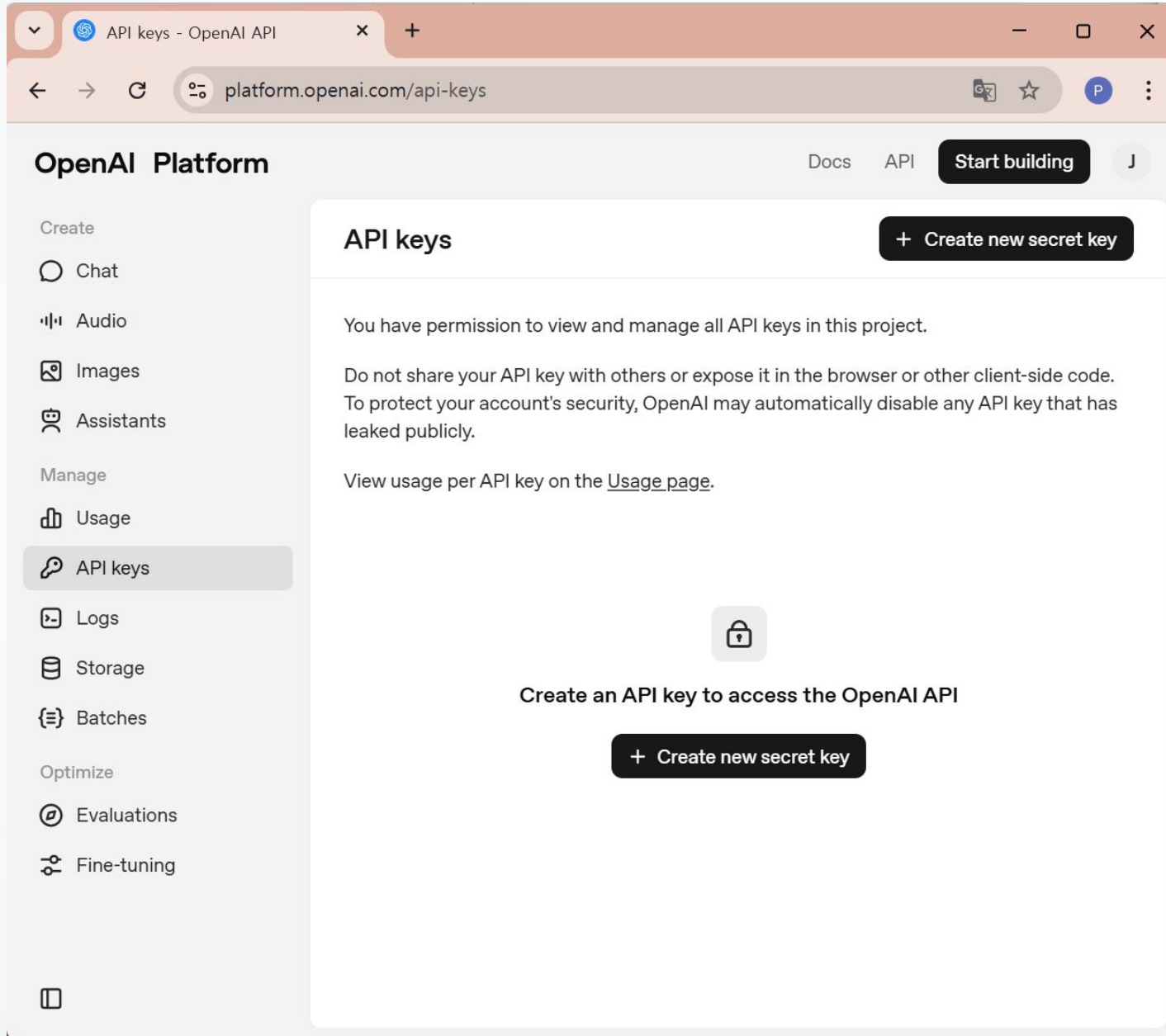
 Google로 계속하기

 Microsoft 계정으로 계속하기

 Apple로 계속하기

 폰으로 계속하기

API Key 발급 - OpenAI(<https://platform.openai.com/api-keys>)



The screenshot shows the OpenAI Platform interface in a web browser. The browser's address bar displays the URL `platform.openai.com/api-keys`. The page header includes the "OpenAI Platform" logo, navigation links for "Docs" and "API", and a "Start building" button. A left-hand sidebar lists various platform features: "Create" (Chat, Audio, Images, Assistants), "Manage" (Usage, API keys, Logs, Storage, Batches), and "Optimize" (Evaluations, Fine-tuning). The "API keys" option in the "Manage" section is highlighted. The main content area is titled "API keys" and contains the following text: "You have permission to view and manage all API keys in this project.", "Do not share your API key with others or expose it in the browser or other client-side code. To protect your account's security, OpenAI may automatically disable any API key that has leaked publicly.", and "View usage per API key on the [Usage page](#)." Below this text is a large lock icon and the instruction "Create an API key to access the OpenAI API". Two buttons, "+ Create new secret key", are visible: one in the top right corner of the main content area and one centered below the lock icon.

API keys - OpenAI API

platform.openai.com/api-keys

OpenAI Platform

Docs API Start building J

Create

- Chat
- Audio
- Images
- Assistants

Manage

- Usage
- API keys
- Logs
- Storage
- Batches

Optimize

- Evaluations
- Fine-tuning

API keys

+ Create new secret key

You have permission to view and manage all API keys in this project.

Do not share your API key with others or expose it in the browser or other client-side code. To protect your account's security, OpenAI may automatically disable any API key that has leaked publicly.

View usage per API key on the [Usage page](#).

Create an API key to access the OpenAI API

+ Create new secret key



API Key 발급 - OpenAI(<https://platform.openai.com/api-keys>)

OpenAI Platform

Docs API Start building J

Create

Chat Audio Images Assistants

Manage

Usage API keys Logs Storage Batches

Optimize

Evaluations Fine-tuning

API keys

+ Create new secret key

Create new secret key

Owned by

You Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

My Test Key

Project

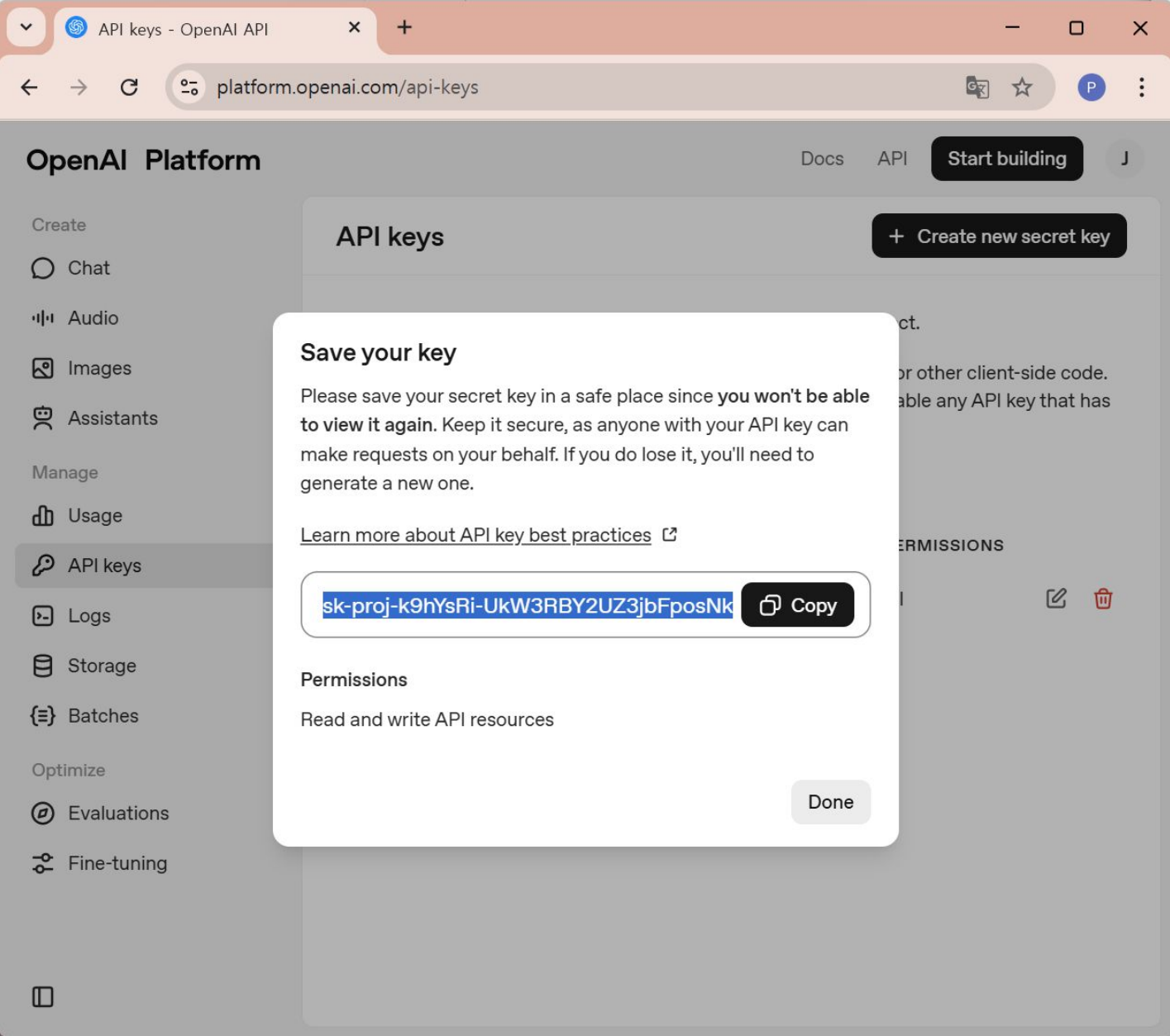
Default project

Permissions

All Restricted Read only

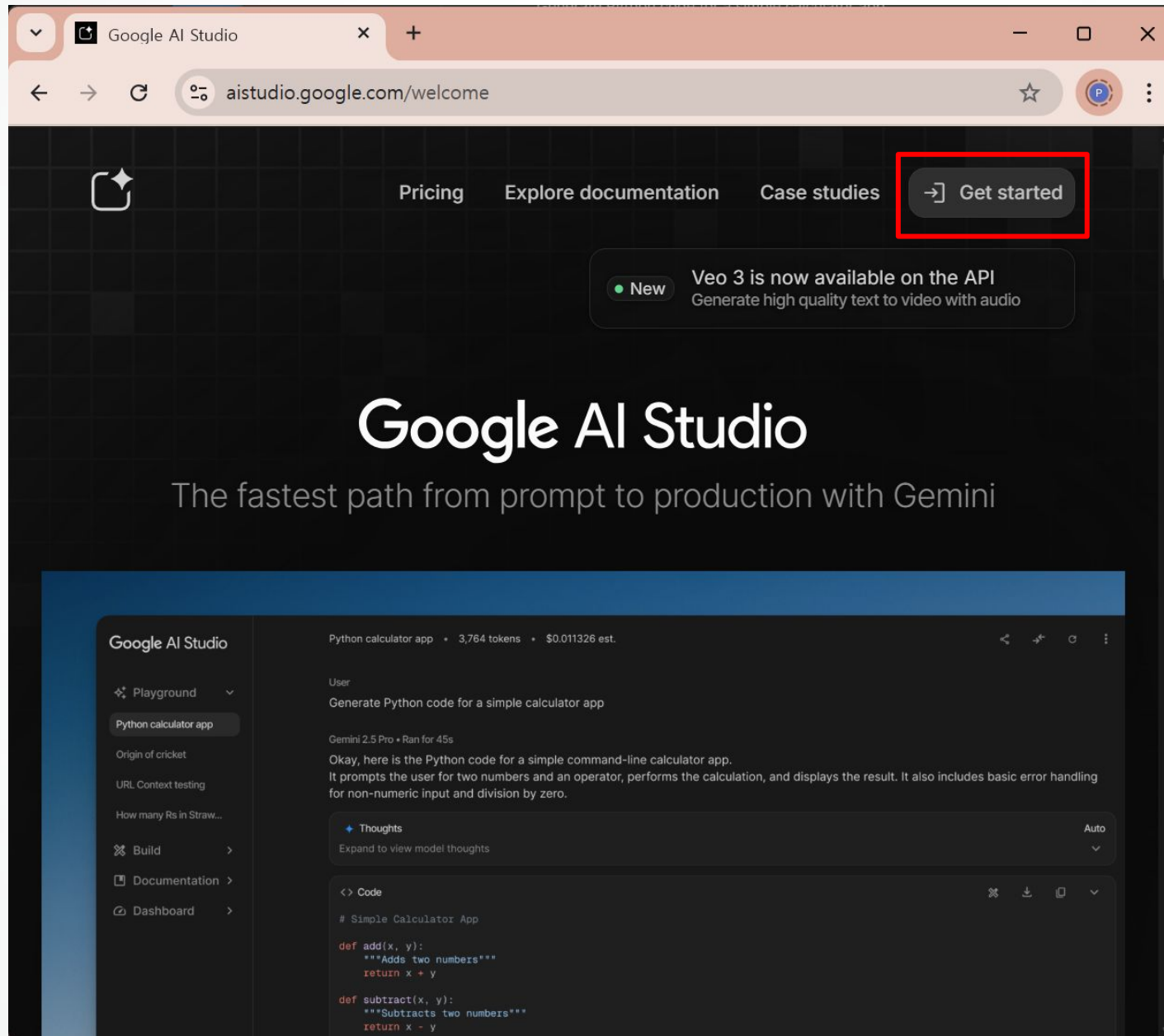
Cancel Create secret key

API Key 발급 - OpenAI(<https://platform.openai.com/api-keys>)



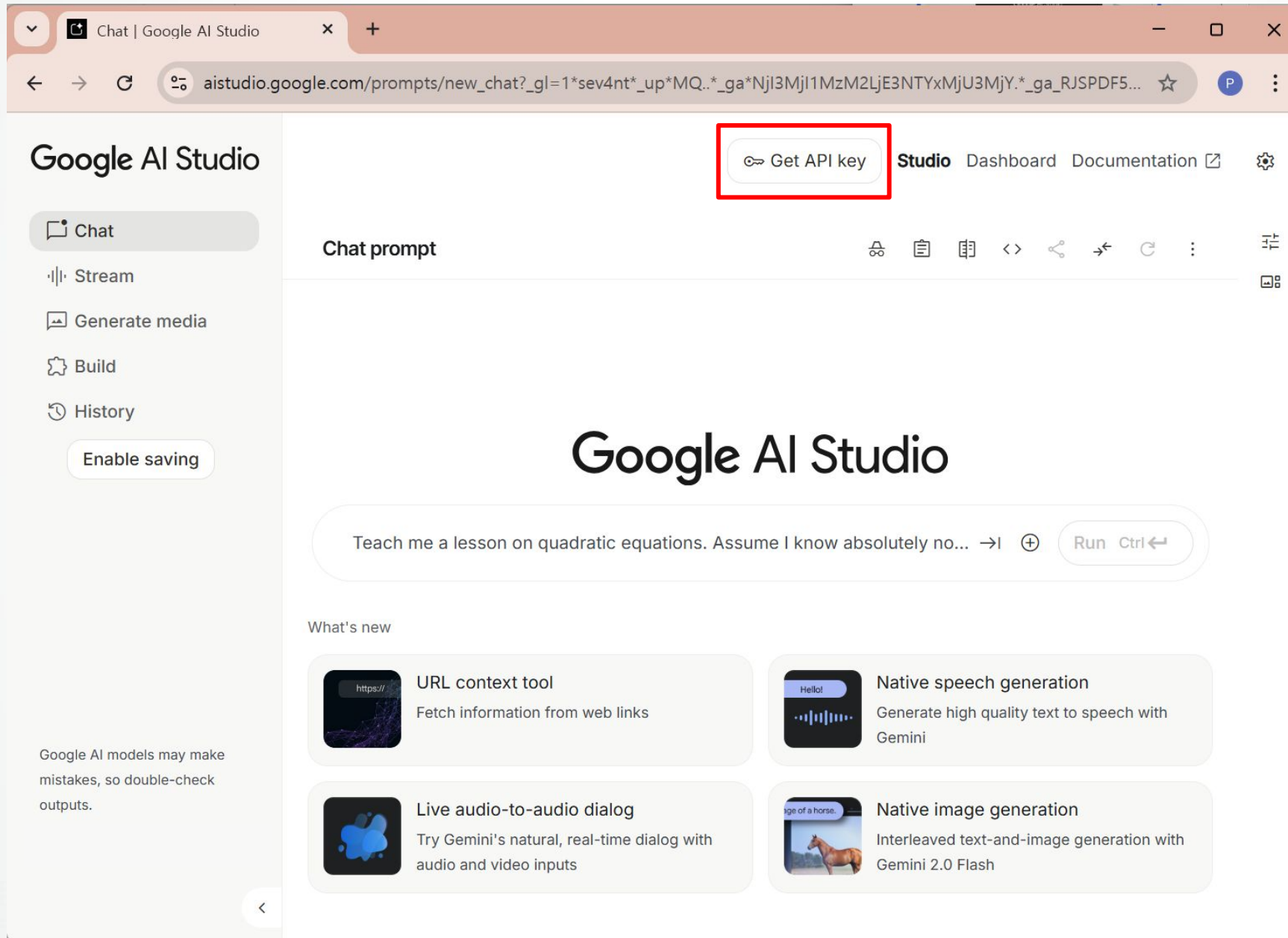
The screenshot shows the OpenAI Platform interface in a web browser. The browser's address bar displays `platform.openai.com/api-keys`. The page title is "API keys - OpenAI API". The main content area is titled "API keys" and features a "+ Create new secret key" button. A modal dialog titled "Save your key" is centered on the screen, containing the following text: "Please save your secret key in a safe place since you won't be able to view it again. Keep it secure, as anyone with your API key can make requests on your behalf. If you do lose it, you'll need to generate a new one." Below this text is a link: "Learn more about API key best practices". A text input field contains the API key `sk-proj-k9hYsRi-UkW3RBY2UZ3jbFposNk`, followed by a "Copy" button. The modal also includes a "Permissions" section with the text "Read and write API resources" and a "Done" button at the bottom right. The background interface includes a sidebar with navigation options like "Create", "Manage", and "Optimize", and a top navigation bar with "Docs", "API", and "Start building" links.

API Key 발급 - Google(<https://aistudio.google.com/welcome>)



로그인 진행

API Key 발급 - Google(<https://aistudio.google.com/welcome>)



The screenshot displays the Google AI Studio web interface. The browser's address bar shows the URL `aistudio.google.com/prompts/new_chat?gl=1*sev4nt*_up*MQ.*_ga*Njl3Mjl1MzZM2LjE3NTYxMjU3MjY.*_ga_RJSPDF5...`. The page features a left sidebar with navigation options: Chat, Stream, Generate media, Build, and History, along with an 'Enable saving' button. The main content area has a top navigation bar with 'Studio', 'Dashboard', and 'Documentation' links. A red rectangular box highlights the 'Get API key' button in the top right corner of the main area. Below this, there is a 'Chat prompt' section with a text input field containing the prompt 'Teach me a lesson on quadratic equations. Assume I know absolutely no...' and a 'Run' button. At the bottom, a 'What's new' section lists four features: 'URL context tool', 'Native speech generation', 'Live audio-to-audio dialog', and 'Native image generation'.

Google AI Studio

Chat

Stream

Generate media

Build

History

Enable saving

Get API key

Studio Dashboard Documentation

Chat prompt

Teach me a lesson on quadratic equations. Assume I know absolutely no... → + Run Ctrl

What's new

URL context tool
Fetch information from web links

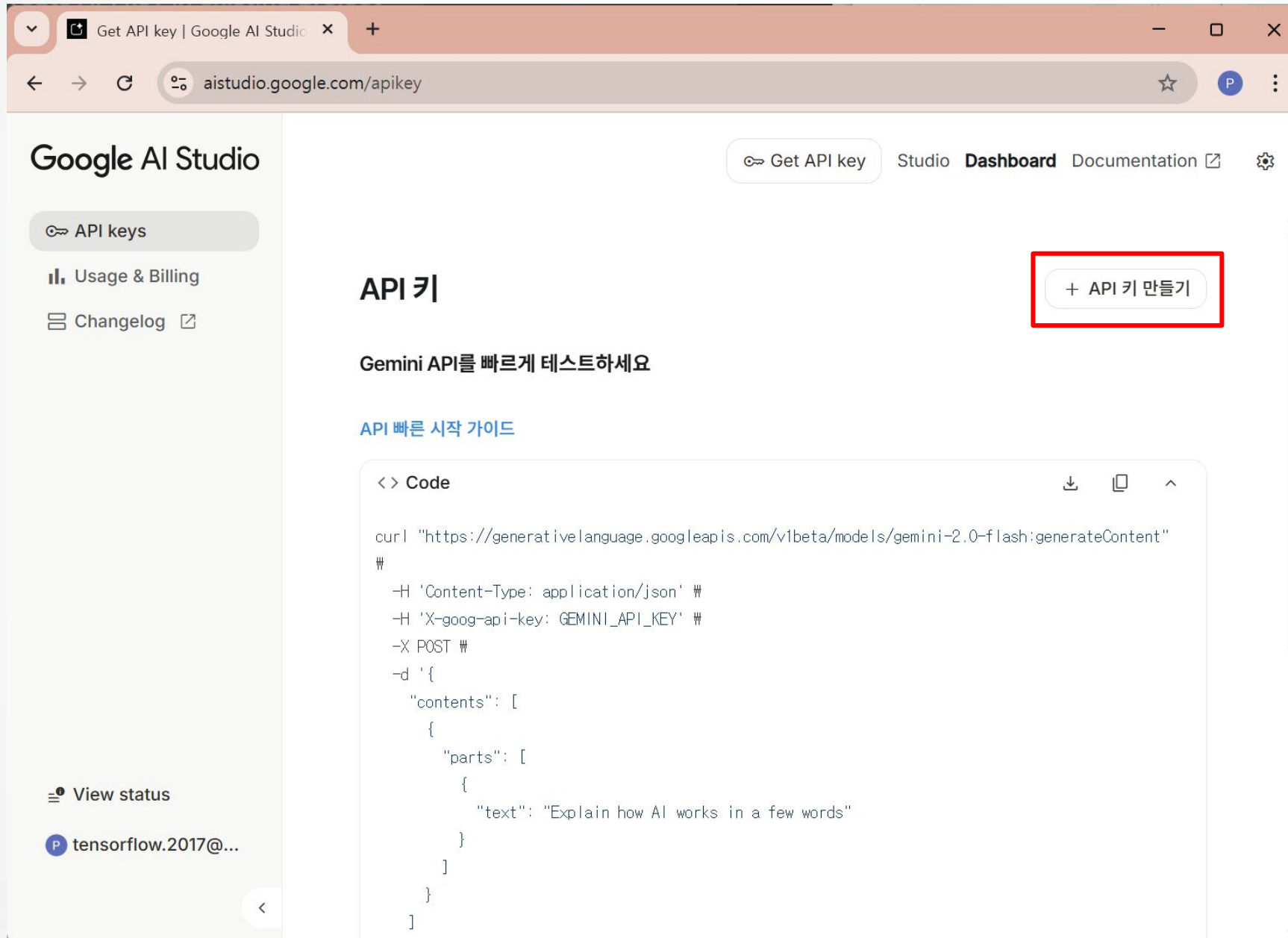
Native speech generation
Generate high quality text to speech with Gemini

Live audio-to-audio dialog
Try Gemini's natural, real-time dialog with audio and video inputs

Native image generation
Interleaved text-and-image generation with Gemini 2.0 Flash

Google AI models may make mistakes, so double-check outputs.

API Key 발급 - Google(<https://aistudio.google.com/welcome>)



Google AI Studio

Get API key | Google AI Studio

aistudio.google.com/apikey

Get API key Studio **Dashboard** Documentation

API keys

Usage & Billing

Changelog

API 키

+ API 키 만들기

Gemini API를 빠르게 테스트하세요

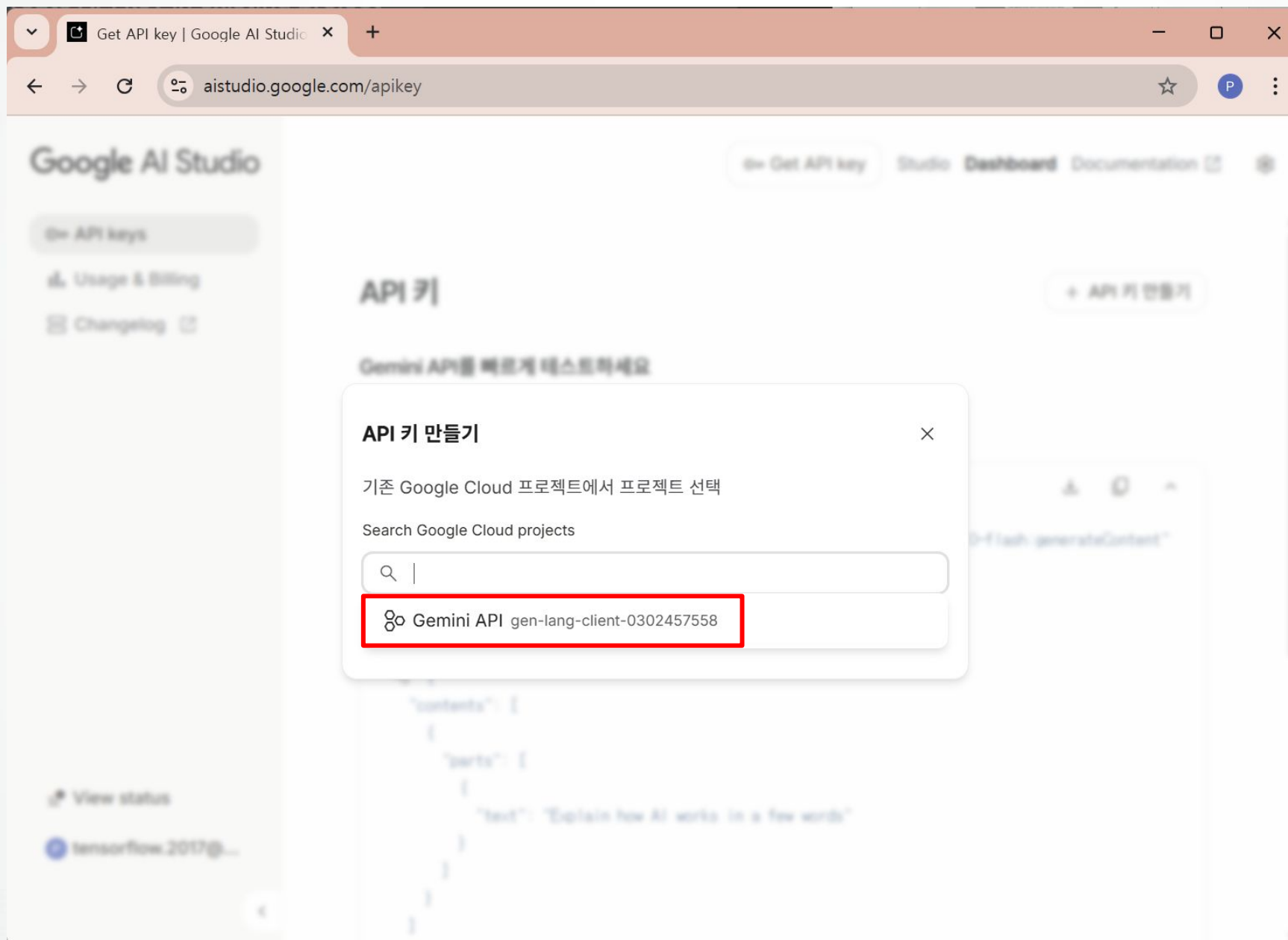
[API 빠른 시작 가이드](#)

```
< > Code
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent"
#
-H 'Content-Type: application/json' #
-H 'X-goog-api-key: GEMINI_API_KEY' #
-X POST #
-d '{
  "contents": [
    {
      "parts": [
        {
          "text": "Explain how AI works in a few words"
        }
      ]
    }
  ]
}
```

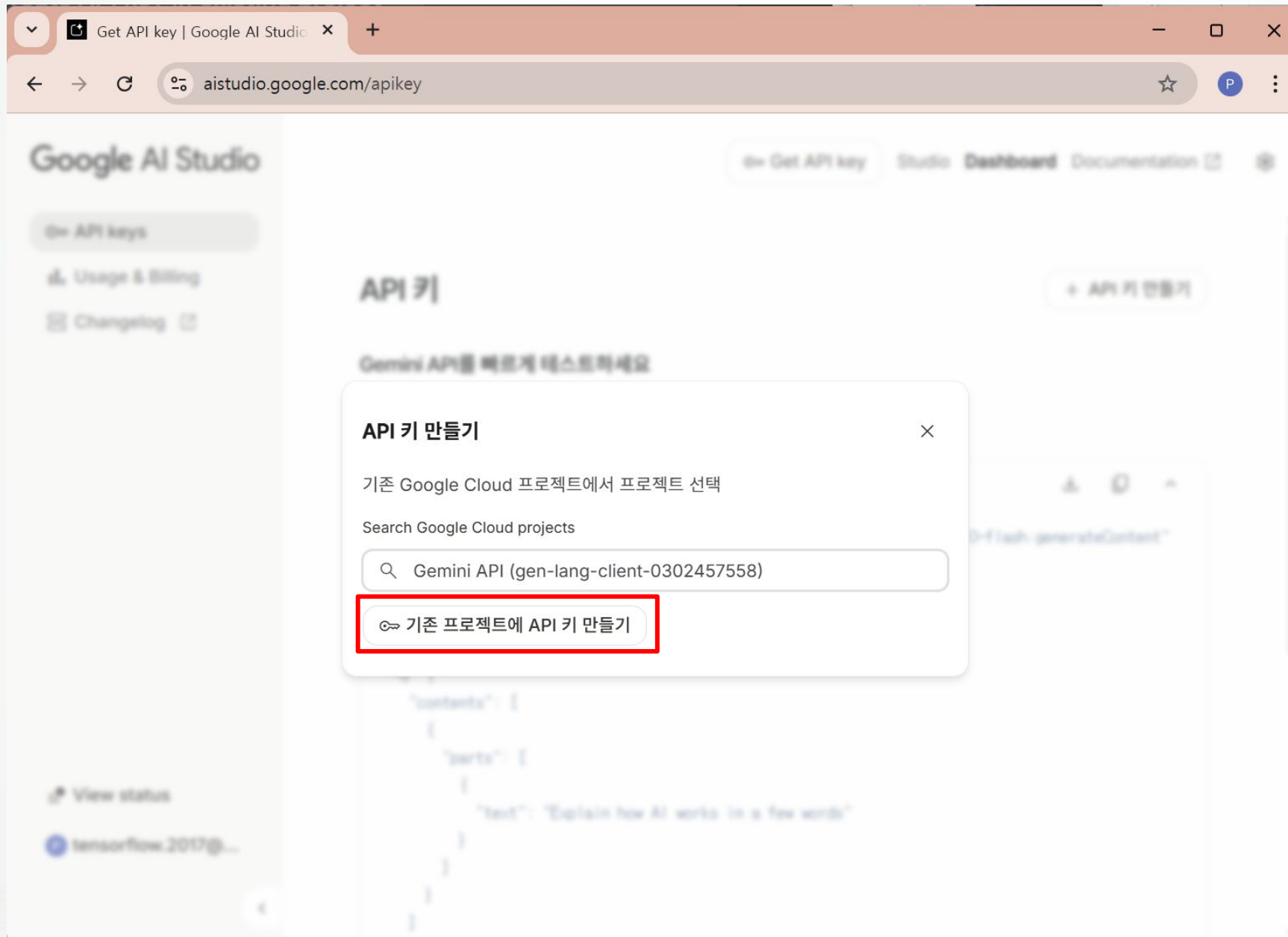
View status

tensorflow.2017@...

API Key 발급 - Google(<https://aistudio.google.com/welcome>)



API Key 발급 - Google(<https://aistudio.google.com/welcome>)





API 키

Gemini API를 빠르게 테스트하세요

API 키를 생성하세요

< Code

API 키는 이

API 키

API 키

API 키

API 키

Get API key

Studio

Dashboard

Documentation

+ API 키 만들기

API 키 생성됨

×

API 키는 안전하게 사용합니다. 다른 사람이 볼 수 있는 코드에 키를 공유하거나 삽입하지 마세요.

복사

API 키

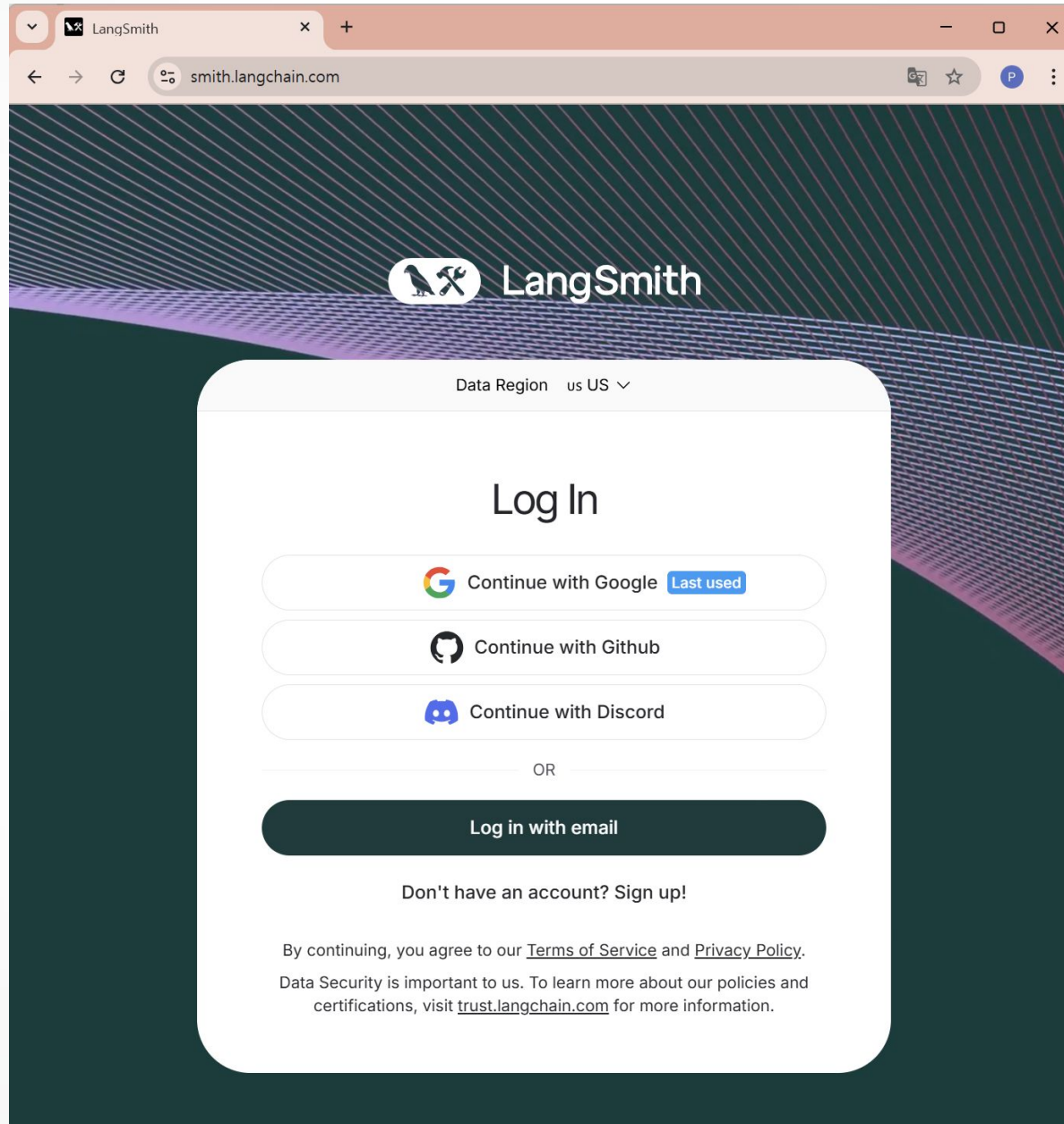
Gemini API

2025. 8. 25.

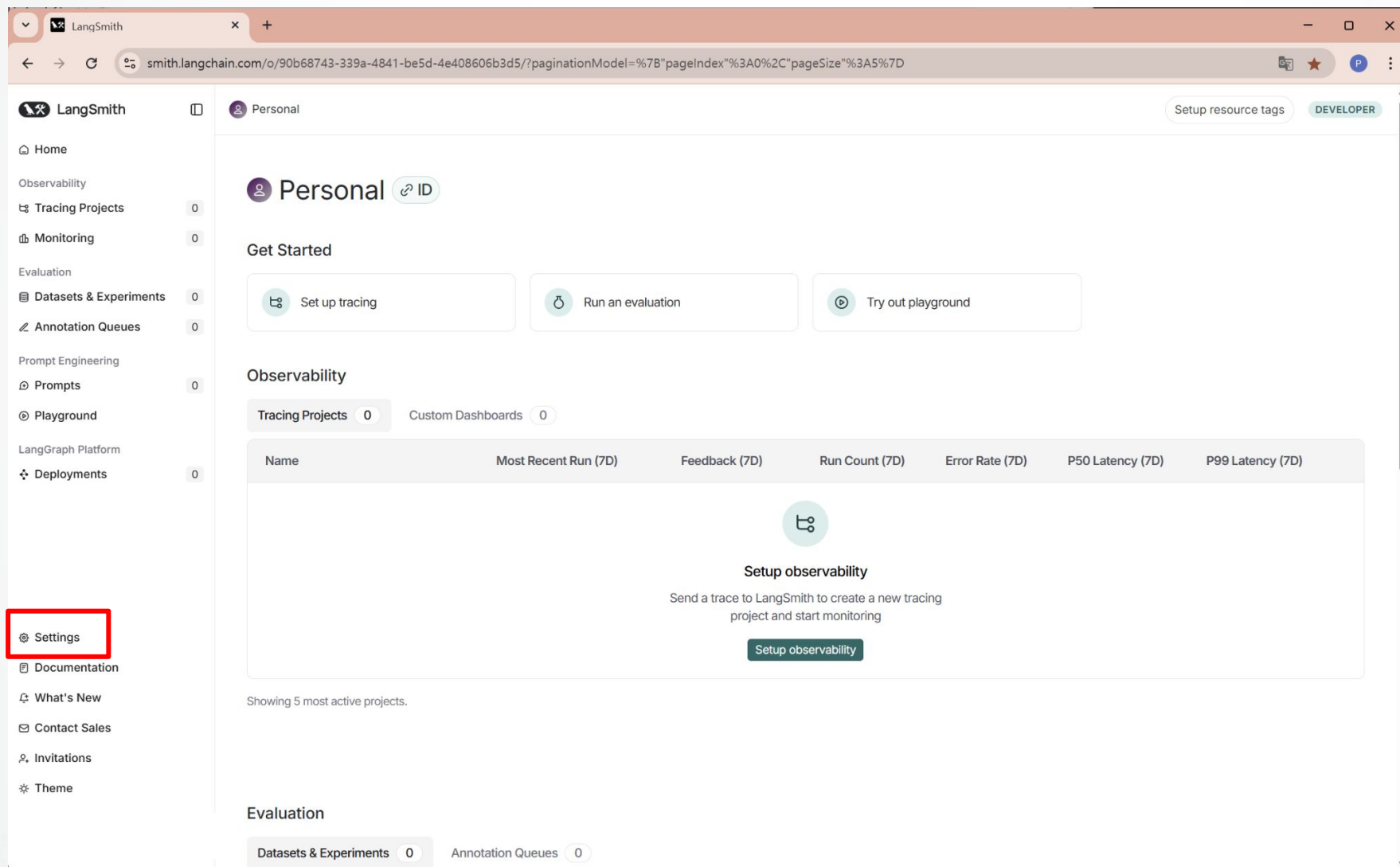
결제 상태
사용 대역폭 보기

API 키는 안전하게 사용해야 합니다. 이러한 API 키를 공개 코드에 공유하거나 삽입하지 마세요. 결제가 사용 설정된 프로젝트에서 Gemini API를 사용하는 경우 [사용한 만큼만 지불하는 가격 책정 방식](#)이 적용됩니다.

API Key 발급 - LangSmith(<https://smith.langchain.com/>)



API Key 발급 - LangSmith(<https://smith.langchain.com/>)



The screenshot displays the LangSmith web application interface. The browser's address bar shows the URL `smith.langchain.com/o/90b68743-339a-4841-be5d-4e408606b3d5/?paginationModel=%7B"pageIndex"%3A0%2C"pageSize"%3A5%7D`. The interface includes a sidebar on the left with the following menu items: Home, Observability, Tracing Projects (0), Monitoring (0), Evaluation, Datasets & Experiments (0), Annotation Queues (0), Prompt Engineering, Prompts (0), Playground, LangGraph Platform, and Deployments (0). The 'Settings' option, located below 'Deployments', is highlighted with a red rectangular box. Other visible options in the sidebar include Documentation, What's New, Contact Sales, Invitations, and Theme. The main content area is titled 'Personal' and features a 'Get Started' section with three buttons: 'Set up tracing', 'Run an evaluation', and 'Try out playground'. Below this is an 'Observability' section with tabs for 'Tracing Projects' (0) and 'Custom Dashboards' (0). A table header is visible with columns: Name, Most Recent Run (7D), Feedback (7D), Run Count (7D), Error Rate (7D), P50 Latency (7D), and P99 Latency (7D). A large button labeled 'Setup observability' is present, accompanied by the text 'Send a trace to LangSmith to create a new tracing project and start monitoring'. At the bottom, an 'Evaluation' section shows tabs for 'Datasets & Experiments' (0) and 'Annotation Queues' (0).

API Key 발급 - LangSmith(<https://smith.langchain.com/>)

LangSmith

smith.langchain.com/o/90b68743-339a-4841-be5d-4e408606b3d5/settings/apikeys

< Back

Organizations Org ID

Personal ▼

Workspaces PLUS

Members and roles PLUS

API Keys

Models

Shared

Secrets

Feedback tags

Rules

Usage and billing

Documentation

What's New

Contact Sales

Invitations

Theme

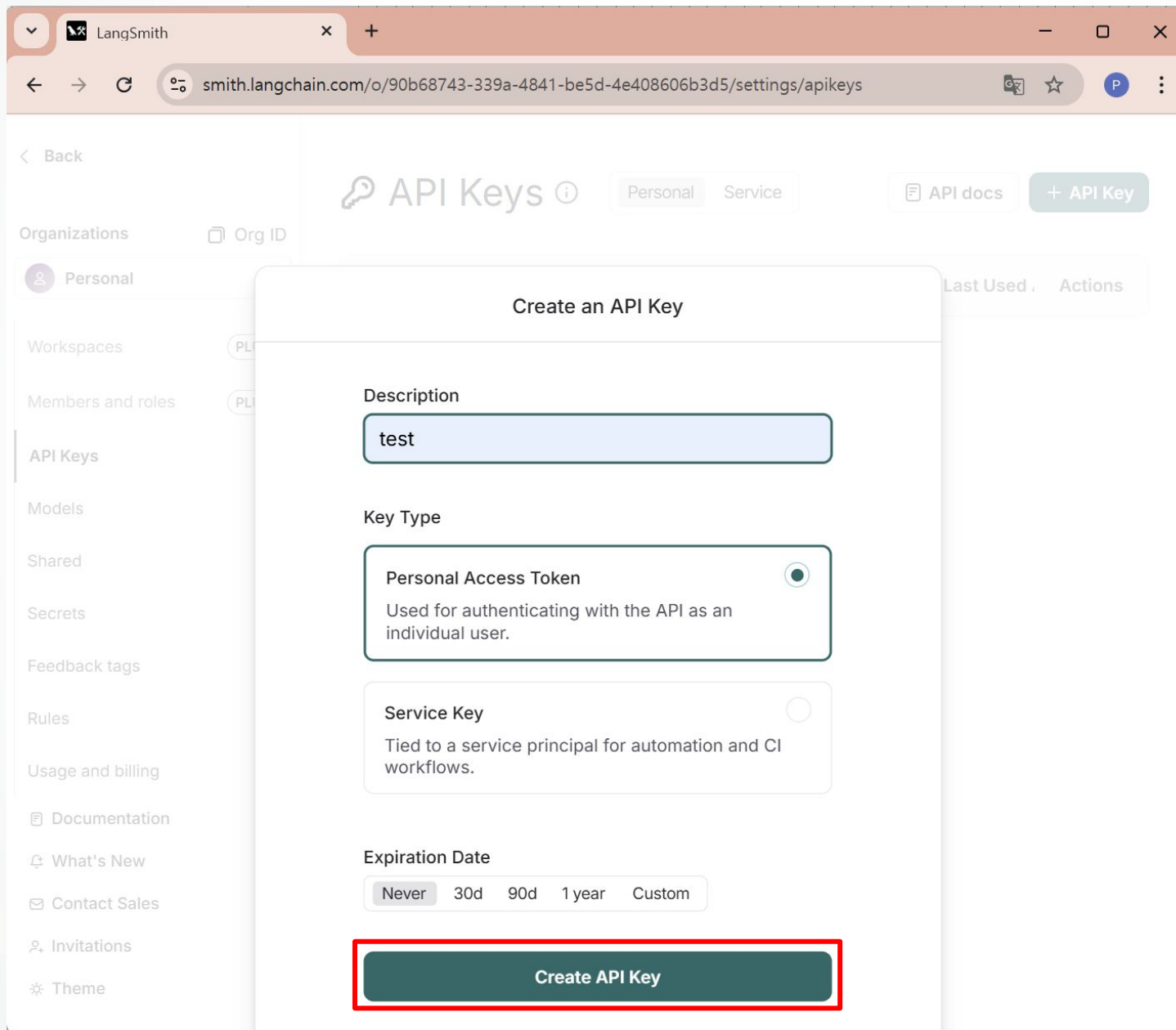
API Keys ⓘ

Personal Service

API docs + API Key

Key	Expires	Description	Created At	Last Used	Actions
-----	---------	-------------	------------	-----------	---------

API Key 발급 - LangSmith(<https://smith.langchain.com/>)



LangSmith

smith.langchain.com/o/90b68743-339a-4841-be5d-4e408606b3d5/settings/apikeys

< Back

API Keys ⓘ

Personal Service

API docs + API Key

Organizations

Org ID

Personal

Workspaces

Members and roles

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Create an API Key

Description

test

Key Type

Personal Access Token
Used for authenticating with the API as an individual user.

Service Key
Tied to a service principal for automation and CI workflows.

Expiration Date

Never 30d 90d 1 year Custom

Create API Key



API Key 발급 - LangSmith(<https://smith.langchain.com/>)

LangSmithsmith.langchain.com/o/90b68743-339a-4841-be5d-4e408606b3d5/settings/apikeys

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API Keys

PersonalService

API docs+ API Key

Key	Expires	Description	Created	Last Us	Actions
lsv2_pt_4 ... 304b	Never	test	20...		

Copy your new API key

Copy

lsv2_pt_4c7baba7b19948bd8f6a10252fc7fbb3_cdi

I've saved the API key to a safe place

LangSmithsmith.langchain.com/o/90b68743-339a-4841-be5d-4e408606b3d5/settings/apikeys

Back

Organizations

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Members and roles

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API Keys

PersonalService

API docs+ API Key

Key	Expires	Description	Created	Last Us	Actions
lsv2_pt_4 ... 304b	Never	test	20...		

◆ LLM (Large Language Model)

● 딥러닝 모델

- 문장 이해/요약, 질문에 답하기, 글 쓰기, 번역하기, 코드 생성
- OpenAI의 GPT, Google의 Gemini, Meta의 LLaMA, Anthropic의

Claude

● 각 회사마다 SDK/API를 제공

- OpenAI: openai (Python SDK)
- Google: google.generativeai (Python/JavaScript/Go/Java SDK)
- Anthropic: REST API + SDK
- Meta: 직접 모델 다운로드/로컬 실행, HuggingFace Transformers

◆ LangChain

● 단순 LLM 호출 (프롬프트 (질문/지시) 를 전달하고 텍스트 형태의 답을 받음) 의 단점

- 문맥 (이전 대화 내용) 반영 불가
- 외부 정보 반영 불가
- 단계를 거쳐서 답을 얻어내는 유형의 작업 처리가 어려움

→ 단순 호출이 아닌 앱을 만들수 있으려면 다양한 기능이 필요

LLM 기반의 다양한 애플리케이션을 만들기 위해서는 다양한 기능을 추가해야 함

◆ LangChain의 특징

- 컴포넌트를 조합해서 전체 앱을 구성
- 다양한 LLM을 하나의 인터페이스로 연결 (모델 교체가 간단)
- 외부 데이터 연결 ; Retrieval Augmented Generation (RAG)
 - PDF/웹 페이지/데이터베이스를 대상으로 검색을 진행한 다음 LLM에 전달
- 도구를 선택해서 작업 수행 (에이전트)
 - API/계산기/브라우저 등을 사용해서 작업 수행
- 메모리 기능
 - 이전 대화 내용 기억/문맥 유지
- 결과 추적/디버깅/성능 비교 목적의 LangSmith 기능 제공
- 앱이나 REST API 형태로 배포가 용이

- ◆ LangChain 컴포넌트
 - LLM
 - Prompt Template
 - Chain
 - Memory
 - Agent
 - Tool
 - Retriever
 - Document Loader
 - Vector Store
 - Callback

